THANGA SRI P

Product Engineer — Full Stack & Product Engineering
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PROFESSIONAL SUMMARY

Product Engineer with 3 years of experience owning a product domain end to end at Hash8.io, a product company building the AEP industrial platform. Primary engineering owner of SIM lifecycle and device connectivity across 2 live production tenants and 950 users — from requirements definition through frontend, backend and delivery. Works directly with product and business stakeholders to define KPIs, and analyzes operational and usage data to identify friction; that analysis drove two platform decisions personally implemented, a schema-driven configuration redesign replacing 25 screens and a multi-tier RBAC access model. Full stack across Angular, React, TypeScript, NestJS and PostgreSQL.

TECHNICAL SKILLS

Frontend: Angular, React, TypeScript, JavaScript (ES6+), RxJS, HTML5, CSS3, Tailwind CSS

Backend: NestJS, Node.js, REST APIs, WebSockets, Authentication & Authorization, RBAC, Multi-Tenant Architecture

Data: PostgreSQL, Neo4j, Temporal, Time-Series Data

Product & Analysis: KPI Definition & Dashboarding, Operational & Usage Data Analysis, Requirements-to-Engineering Translation, Stakeholder Communication

Visualization: ApexCharts, D3.js, Real-Time Dashboards

Tools & Practices: Git, GitHub, CI/CD, Docker, Agile/Scrum, PR Review, GitHub Copilot, Claude AI, MCP

PROFESSIONAL EXPERIENCE

IoT Platform Engineer

Aug 2023 – Present

Hash8.io Pvt. Ltd., Hosur, Tamil Nadu — Product-Based Company, AEP Industrial Platform

Product Ownership

- Own the SIM lifecycle and device connectivity domain end to end as primary engineering owner — activation, suspension, private APN exposure, device provisioning and asset onboarding — across 2 live production tenants and 950 users.
- Translate business requirements into engineering specifications in the Functional Working Group, and communicate delivery status and technical decisions to senior stakeholders including CEO-level leadership.
- Hold sprint-level delivery ownership for the connectivity domain; govern branching strategy and lead PR review for its codebase.

Data-Informed Product Decisions

- Analyzed device connectivity data and dashboard usage patterns to identify configuration gaps and workflow friction points, directly informing two platform redesigns.
- Replaced 25 hardcoded configuration screens with a schema-driven Angular Formly engine as a direct result of that analysis, converting new asset-type onboarding from a development task into a configuration change.
- Designed and implemented a multi-tier RBAC model, enforced at both the UI and API layer, delivering tenant- and role-scoped data visibility across analytics and monitoring products.
- Built KPI dashboards defined with product and business stakeholders — device uptime, SIM activation status, telemetry health, active device counts, tenant usage and feature adoption.

Full Stack Delivery

- Delivered connectivity features across frontend and backend in Angular and NestJS over PostgreSQL, supporting fleets of ~2,000 and ~100,000 provisioned devices.
- Integrated the SIMETRY MVNO carrier platform, designing a two-source-of-truth architecture bridging carrier-owned SIM state with OEM-owned device mapping via asset-ID synchronization.
- Engineered Temporal durable workflows for 30–60 second asynchronous carrier operations, surviving worker restarts and deployments with bounded retries.
- Implemented WebSocket push for real-time operation status, removing client-side polling so users see accurate long-running operation state without refreshing.

Product Surface Breadth

- Built frontend across four AssetiQ monitoring applications — Waste Water Monitoring, Reefer Tracking, Water Management and Tank Monitoring — covering live status, historical views, map visualization, command interfaces and configuration.
- Designed a shared, reusable UI component framework and navigation architecture adopted across all AssetiQ applications, accelerating delivery of each new application.
- Reduced full page load to under 500ms on real-time telemetry dashboards through OnPush change detection, trackBy row reuse and virtualized rendering.

KEY PROJECTS

Schema-Driven Configuration Redesign

Tech: Angular, Angular Formly, TypeScript

- **Problem:** usage analysis showed configuration screens were a recurring friction point, and each new asset type required its own purpose-built screen.
- **Decision:** replace per-screen development with a single metadata-driven engine, trading some per-screen flexibility for delivery speed.
- **Outcome:** 25 hardcoded screens replaced by one engine; new asset-type onboarding became a configuration change.

SIM Lifecycle & Carrier Connectivity Platform

Tech: Angular, React, TypeScript, NestJS, PostgreSQL, Temporal, WebSockets

- **Problem:** customers needed to activate, suspend and provision SIMs for field equipment, but authoritative state lived across two external systems.
- **Architecture:** two-source-of-truth model bridged by asset-ID sync, Temporal durable workflows for async carrier operations, WebSocket push for live status.
- **Scale & outcome:** 2 production tenants, ~2,000 and ~100,000 provisioned devices, 950 users, with lifecycle operations surviving deployments and restarts.

EDUCATION

B.Sc. Computer Science | 2020 – 2023 | CGPA: 7.9

S.T. Hindu College, Manonmaniam Sundaranar University, Nagercoil